

Dongho Kim

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Education

Northwestern University

M.S. in Statistics and Data Science

University of California, San Diego

Exchange Student, Business Economics and Data Science

Dongguk University

B.B.A. in Business Administration; B.S. in Data Science Software

Evanston, IL

Sep 2025 – Jun 2027

San Diego, CA

Sep 2022 – Jun 2023

Seoul, South Korea

Mar 2019 – Feb 2025

Research Interests

Reliable agentic AI for expert workflows: building verification-first systems that route evidence, audit unsupported claims, and measure uncertainty and failure modes in AI-assisted writing and computational social science.

Research Experience

Graduate Research Intern

MAGICS Lab, Northwestern University; Advisor: Prof. Han Liu

Sep 2025 – Present

Evanston, IL

- Co-built SPINE's validation-gated robot-debugging loop: robot profiles and diagnostic subagents propose fixes, safe tools apply them, and live teleoperation probes decide when a case can close.
- Devised blind compounded-fault stress tests on DOBOT X-Trainer and AgileX PiPER, where software, hardware, and hybrid bugs mask one another and partial fixes leave hidden failures.
- Showed validation-gated closure outperformed direct human-LLM debugging: SPINE resolved every implanted fault (DOBOT 14/14; PiPER 10/10) and raised DOBOT recovery success from 75% to 100%.

Research Collaborator

NSF-Simons SkAI Institute, Northwestern University

Oct 2025 – Present

Evanston, IL

- Evaluated foundation-model representations in StarEmbed, an ICML 2026 benchmark of roughly 40k expert-labeled astronomical light curves.
- Identified reliability limits under irregular sampling, heteroskedastic noise, and out-of-distribution shifts in scientific time-series transfer.

Research Assistant

Intelligence and Optimization in Networks Lab; Advisor: Prof. Hyeryung Jang

Dec 2023 – Aug 2024

Seoul, South Korea

- Built hybrid RNN-sentiment models for stock-return prediction across 88 firms.
- Added FinBERT sentiment features to a forecasting pipeline, improving predictive accuracy by 51%.

Research Projects

Sci2Pol-Agent: Verification-First Agentic Harness for Science-to-Policy Writing

2026

Northwestern University

- Built a training-free verification harness that routes a fixed language model through evidence extraction, support checking, local revision, and format control.
- Improved Sci2Pol-Bench overall score by +2.6 points under a same-backbone evaluation, with strongest gains in Verification and documented failure/cost traces.

Information Contagion and Market Connectivity

Northwestern University, Computational Social Science

Sep 2025 – Dec 2025

- Modeled NASDAQ firm co-mention networks with multilevel and ERGM methods to study how information diffusion relates to market outcomes.

Publications and Preprints (* equal contribution)

- Ham, M.*, **Kim, D.***, Lee, C.*, Wang, J.*, Zhao, J., Ye, G., Park, S., Kim, M. J., Zhang, Y., and Liu, H. *SPINE: Bridging the Cyber-Physical Gap with Agentic AI*. Preprint manuscript, 2026.
- Li, W.*, Chen, H.-Y.*, Rehemtulla, N.*, Shah, V., **Kim, D.**, Wu, D., Lin, Q., Miller, A., and Liu, H. *StarEmbed: Benchmarking Time Series Foundation Models on Astronomical Observations of Variable Stars*. ICML 2026; icml.cc/virtual/2026/poster/63677; arXiv:2510.06200.

Professional and Teaching Experience

Teaching Assistant

Sep 2023 – Jun 2024

Department of Business Administration, Dongguk University

Seoul, South Korea

- Built Excel-based grading and reporting tools for Human Resource and Diversity Management courses.
- Generated course performance reports that supported a 95% student satisfaction rating.

Operations Assistant

Nov 2021 – Mar 2022

M&E Parts

Seoul, South Korea

- Developed an inventory analytics system to monitor stock levels and reduce supply-management errors.
- Analyzed transaction records and reduced excess inventory by 10% through data-driven process monitoring.

Sergeant, Military Service

Jan 2020 – Sep 2021

Republic of Korea Ministry of National Defense

South Korea

- Led supply management for mission-critical communication equipment and maintained operational readiness documentation.
- Implemented process controls that reduced reporting errors by 30%.

Technical Skills

Programming: Python, SQL, Git, LaTeX; Pandas, NumPy, Scikit-learn, PyTorch

Machine Learning: LLM agent systems, verification harnesses, evaluation and benchmarking, time-series foundation models, NLP

Statistics and Data: Regression, hypothesis testing, feature engineering, experimental design, network analysis

Visualization: Matplotlib, Seaborn, exploratory data analysis

Awards and Scholarships

- Excellence Award, Seoul City AI Video Content Competition, 2024
- Top 3%, LG AI Research B2B Sales Opportunity Prediction Competition, 2024
- Exchange Program Scholarship, Dongguk University, 2023
- Academic Excellence Scholarship, Dongguk University, 2022

Service and Languages

- International Student Mentor, International Buddy Program, Dongguk University, 2019
- Korean: native; English: professional working proficiency; Chinese: intermediate (HSK Level 4)